

AI-POWERED CREDIBILITY INTELLIGENCE

TruthScan

Fake News Detection using NLP & AI

- NLP ANALYSIS
- CLAUDE AI
- ZERO DATA RETENTION

SUBMITTED BY

Vyshnavi

B.Tech CSE (Data Science)
Dhanekula Institute of Engineering & Technology
Vijayawada, Andhra Pradesh



FEATURES

- Instant NLP signal analysis
- Claude AI deep review
- Confidence score gauge
- Session history & analytics
- Export reports (Copy / JSON)
- Dark & light mode
- Fully responsive design
- Zero server — fully private

Problem Statement

3.8B

Social media users exposed
to misinformation daily

76%

People unable to identify
fake news consistently

70%

Faster spread of fake news
vs real news (MIT Study)

CURRENT CHALLENGES



Manual fact-checking is SLOW

Takes hours to days; news spreads in seconds



Existing tools lack TRANSPARENCY

No explanation of why an article is flagged



No REAL-TIME solution

Readers need instant verification while reading



PRIVACY concerns

Many tools upload article text to external servers

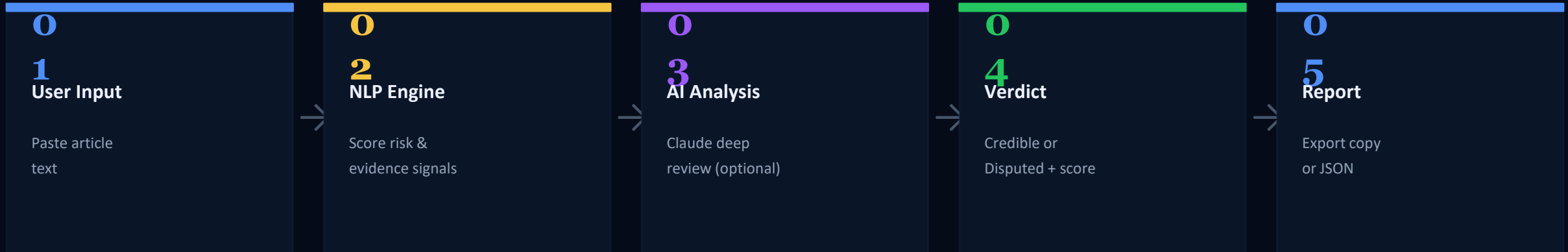
Existing Systems & Limitations

Tool / Method	Speed	Transparency	Privacy	Accuracy	Availability
Snopes / PolitiFact	✗ Slow (hours)	✓ Good	✓ Good	✓ High	✗ Manual
Google Fact Check	⚠ Medium	⚠ Partial	✓ Good	⚠ Medium	⚠ Limited
ML/BERT Models	✓ Fast	✗ Blackbox	✗ Data sent	✓ High	✗ Technical
Browser Extensions	✓ Instant	✗ None	⚠ Varies	✗ Low	⚠ Limited
TruthScan (ours)	✓ Instant	✓ Full	✓ Zero	⚠ Heuristic	✓ Free / Open

Our Advantage

TruthScan is the only solution that combines instant analysis, full explainability, and complete privacy in one free, open browser app.

Our Proposed Solution



KEY INNOVATIONS



Zero Privacy Risk

Runs 100% in browser — no article text ever sent to a server (unless user opts in to AI)



Explainability

Every verdict includes a linguistic justification — users understand WHY, not just WHAT



Instant Analysis

NLP analysis runs in <1ms on device; no waiting for a server round-trip



Optional AI Depth

Claude AI provides expert-level contextual review beyond pattern matching

Project Objectives

01 Instant Detection

Provide real-time fake news classification in under 1 second, directly in the browser without any server dependency.

02 Explainable AI

Ensure every verdict includes a clear explanation of which linguistic signals influenced the classification.

03 Privacy-First Design

Process all article text locally — no user data transmitted to external servers without explicit consent.

04 AI Augmentation

Integrate Claude AI for deep contextual analysis beyond pattern matching, available as an optional enhancement.

05 Professional UX

Build a polished, responsive, accessible interface suitable for journalists, students, and general public.

06 Open & Free Access

Deliver the tool as a single-file HTML application deployable on GitHub Pages — free, open, zero infrastructure cost.

Technologies Used

FRONTEND

- **HTML5**
Semantic structure, ARIA accessibility, PWA manifest
- **CSS3**
Custom properties, Grid, Flexbox, animations, glassmorphism
- **JavaScript ES6+**
Modules, async/await, DOM APIs, sessionStorage

DESIGN

- **Playfair Display**
Editorial serif font for authoritative headings
- **Inter**
Clean sans-serif for body text and UI elements
- **JetBrains Mono**
Monospace for technical labels and data values

AI & NLP

- **Claude API**
claude-sonnet-4-6 model for deep credibility analysis
- **NLP Heuristics**
Weighted keyword scoring, Flesch-Kincaid readability
- **Pattern Matching**
Regex-based feature extraction for linguistic signals

DEPLOYMENT

- **GitHub Pages**
Free static hosting for the single-file HTML app
- **SessionStorage**
Client-side temporary storage, zero backend
- **Zero Dependencies**
No npm packages — pure HTML/CSS/JS

System Architecture

BROWSER ENVIRONMENT (CLIENT-SIDE ONLY)

UI LAYER

Input Panel

Result Panel

History Strip

Analytics Cards



SUPPORT MODULES

Toast System

History Manager

Analytics Manager

Theme Manager

Export System



ANALYSIS ENGINE

NLP Heuristics (Risk Score)

Evidence Scoring

FK Reading Level

Feature Extraction

EXTERNAL (OPTIONAL - USER CONSENT)

Anthropic Claude API

claude-sonnet-4-6 | max_tokens: 400 | User must click "Deep AI" to trigger

⚠ Only called on explicit user action — no passive data transmission

Modules & Features



Analysis Engine

- Sensational word detection (29 phrases)
- Evidence signal scoring (26 patterns)
- Attribution phrase recognition (10 patterns)
- ALL-CAPS and punctuation analysis
- Flesch-Kincaid reading level



Result Renderer

- Animated SVG radial confidence gauge
- Linear progress meter with dot indicator
- Risk vs Evidence cue cards
- Animated feature breakdown bars
- Natural language verdict explanation



AI Deep Analysis

- Optional Claude API integration
- Expert credibility assessment prompt
- 3-4 sentence actionable review
- Animated loading indicator
- Graceful API failure fallback



History & Analytics

- Session history (last 12 analyses)
- Clickable history reload
- Total / Real / Fake counters
- Average confidence tracker
- Animated analytics bar charts



Export System

- Clipboard copy (formatted text report)
- JSON download with metadata
- Timestamped filenames
- One-click, no dialogs



UX Systems

- Dark / Light mode toggle
- Toast notification system
- Skeleton loading state
- Onboarding overlay
- Keyboard shortcut (Ctrl+Enter)

Application UI Overview

INPUT PANEL — DARK MODE

[Article Textarea]

[Real / Fake]

[Analyze]

Gauge

VERDICT

Confidence Score

[Feature Breakdown]

RESULT PANEL — CREDIBLE

[Article Textarea]

[Real / Fake]

[Analyze]

Gauge

✓ CREDIBLE

Confidence Score

82%

[Feature Breakdown]

RESULT PANEL — DISPUTED

[Article Textarea]

[Real / Fake]

[Analyze]

Gauge

✗ DISPUTED

Confidence Score

91%

[Feature Breakdown]

ADDITIONAL SCREENS

History Strip

Analytics Dashboard

Light Mode

Onboarding Overlay

Mobile View

Algorithm Deep Dive

SCORING FORMULAS

```
Risk Score  = (sensational_hits × 16) + (exclamations × 6) + (all_caps_words × 8) + (questions × 3)

Evid Score  = (evidence_hits × 13) + (attribution_hits × 11) + (has_numbers ? 15 : 0) + clamp(word_count / 8, 0, 20)

Verdict     = evidScore ≥ riskScore → "Credible"   |   evidScore < riskScore → "Disputed"

Confidence  = clamp(50 + |evidScore - riskScore| × 0.55, 42, 96)
```

RISK SIGNALS (Fake Indicators)

- ⚠ "shocking","you won't believe","miracle"
- ⚠ "secret remedy","hate this trick","exposed"
- ⚠ "share now","hidden for decades","bombshell"
- ⚠ "mainstream media","deep state","BREAKING:"
- ⚠ Excessive !! marks | ALL-CAPS words (3+ chars)

EVIDENCE SIGNALS (Real Indicators)

- ✓ "according to","published","peer-reviewed"
- ✓ "Dr.,""professor","scientists","researchers"
- ✓ "percent","%","statistics","data shows"
- ✓ "official","confirmed","spokesperson"
- ✓ Numeric data present | Text length bonus

Future Scope

PHASE 1 - NEAR TERM (3-6 MONTHS)

- URL-based article fetching — auto-extract text from any news URL
- Browser extension — Chrome/Firefox for in-context analysis
- Multi-language support — Hindi and Telugu for regional news
- Export to PDF — printable credibility report

PHASE 2 - MEDIUM TERM (6-12 MONTHS)

- ML model integration — DistilBERT fine-tuned on FakeNewsNet dataset
- User authentication — saved analysis history across sessions
- REST API backend — batch analysis for institutions and researchers
- Mobile app — React Native iOS and Android application

PHASE 3 - LONG TERM (1-2 YEARS)

- Real-time social media monitoring — Twitter/X, Facebook integration
- Multi-model ensemble — combine BERT, GPT, and Claude for higher accuracy
- Institutional dashboard — for newsrooms and fact-checking organizations
- Academic research tool — bias detection, source reliability scoring

Conclusion

TruthScan demonstrates that a professional, AI-augmented fake news detection tool can be built as a zero-dependency browser application — instant, private, explainable, and freely accessible to everyone.

KEY ACHIEVEMENTS



Professional UI

Glassmorphism, animations, responsive, dark/light mode



Full Explainability

Every verdict shows which signals influenced the result



AI Integration

Claude API provides expert-level deep analysis on demand



Privacy-First

Zero data transmitted without explicit user consent



Production Security

XSS prevention, input validation, no data retention



Complete Feature Set

History, analytics, export, keyboard shortcuts, PWA manifest

Thank You